The Electricity Forward Premium: Renewable Energy Sources and Skewness Preferences

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Abstract

There is mixed empirical evidence on the premiums that explain electricity forward prices. We argue that the growth of electricity supplied by renewable energy sources and the preference for skewness may provide an explanation. We extend an existing equilibrium model by incorporating these features. The model predicts that the influence of spot price variance and skewness on forward prices is undetermined, reconciling mixed empirical findings. In addition, we find additional premiums, related to the covariance and coskewness of renewable supply and spot prices, that explain the forward premium. We provide empirical support using data from the German power market.

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1 Introduction

Since the end of the 1990's, two major changes have taken place in worldwide power markets. The first change was the liberalization of power markets, such that power prices are determined by demand and supply in competitive markets. The second change is the energy transition. While power was predominantly produced by fossil fuels in many countries, these conventional energy sources (CES) face increasing competition from renewable energy sources (RES) such as wind and solar. Although CES and RES produce the same good (power), their economic characteristics differ, which affects the power price in two ways.

First, CES have positive marginal production costs as they burn fuels and need emission rights to produce one additional unit of power. RES have negligible marginal production costs as they only require more wind or solar radiation to produce additional power. Therefore, the growth of RES lowers market prices ceteris paribus. Second, the CES supply is non-intermittent and can be adjusted flexibly, while RES supply is intermittent as output depends on varying wind and solar radiation conditions.[1] Hence, the growth of renewable energy leads to increased price volatility when power storage capacity is not sufficient and/or power demand does not follow the RES supply.

Agents in power markets use forward and future contracts that involve the delivery of power during a future time period to manage power price volatility. As long as power storage capacity and demand response are not sufficient to counterbalance the uncertainty in demand and supply from RES, the theory of storage cannot be applied to price power forward contracts.[2] Instead, researchers follow the expectation theory that relates power forward prices to expected future spot prices and forward premiums. An important paper in this respect is Bessembinder and Lemmon (2002) (hereafter: BL). They propose an equilibrium forward price model for markets where power is produced by CES, demand is uncertain, and storage capacity is negligible. They show that the forward price equals the expected average spot price during the delivery period plus two premiums that are negatively related to the variance and positively related to the skewness of spot prices. However, we no longer observe this in the recent German power market. As shown in Table 1, the skewness and variance of spot prices weakly explain the forward premium in both the baseload and

[1]Therefore, we limit ourselves to intermittent RES, although other types of RES exist, such as non-intermittent hydro power stations.

[2]The theory of storage assumes that a good can be stored and carried from one period to another.

peakload periods, and the signs of estimates are not aligned with the BL prediction.[3]

Table 1: **BL Model Results**

This table presents the BL model results for the ex-post forward premium in baseload and peakload contracts. The results are obtained by data observed on the last trading days of the months between January 2015 and April 2021. HAC standard errors are reported in parentheses.

	Baseload	Peakload
Var(p_S)	0.12	-0.07^{*}
	(0.12)	(0.03)
Skew(p_S)	0.01	-0.03
	(0.06)	(0.02)
R^2	0.04	0.08
Adj. R^2	0.02	0.05
N	76	76

$^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

As the energy transition changes power price dynamics, we are motivated to revisit their work and examine the extent to which forward price formation changes when RES plays a role in the power system. We are not alone. Several studies have extended or tested the BL model. Bühler and Müller-Merbach (2009) develop a generalized dynamic version of the BL model. Oliveira and Ruiz (2021) add market power and production cost uncertainty. Ullrich (2007) adds capacity constraints to the model. Gianfreda et al. (2022) derive an equilibrium model that retains higher-order moments beyond variance and skewness in a Taylor expansion. Koolen et al. (2021) show the impact of different technologies on spot and forward prices. Schwenen and Neuhoff (2024) come close and derive forward prices from a framework that includes RES supply uncertainty. However, their approach is limited as they do not allow for demand uncertainty, and they assume flat (not convex) CES marginal production. They find that the covariance between power spot prices and supply from RES can also explain the forward premiums other than the variance and skewness of spot prices.

Others have empirically examined what factors explain forward premiums. Longstaff and Wang (2004) examine forward premiums as the difference between day-ahead prices and intraday

[3] In power markets, baseload refers to continuous supply across all hours of the day, while peak load denotes delivery during high-demand periods (typically weekdays, 8 AM–8 PM), with peak prices generally higher due to production relying more on costly CES. Off-peak hours are simply the baseload hours, excluding the defined peak period.

spot prices in the PJM market from 2000 to 2002 and find results supporting BL. Haugom and Ullrich (2012) analyze a longer time period and find that the coefficients for variance and skewness vary over time. They do not find evidence of significant forward premiums overall. Viehmann (2011) finds support for the predictions of BL in the German power markets. Douglas and Popova (2008) observe that, although power cannot be stored (yet), we can store fuels with which power is produced such as natural gas. They show that higher gas inventory levels coincide with lower forward premiums. Huisman and Kilic (2012) use this idea of the storability of underlying fossil fuels to show that forward premiums depend on whether the markets predominantly use supply from CES or RES. Lucia and Torró (2011) examine data from Nordic markets wherein hydropower is a dominant form of RES. They find evidence for time-varying forward premiums: positive in winter when low reservoir levels make supply risk high, but zero premiums in summer when water is abundant and risk is low. Redl and Bunn (2013) show that reserve margin,[4] market power,[5] and underlying fuel price volatility all significantly influence the forward premium. Bunn and Chen (2013) find that the main drivers of peak forward premiums are lagged premiums and prices. They find that forward premiums positively relate to spot price volatility and negatively to skewness in the British power market, being the opposite of what BL predict.

We are aware of the mixed results in the literature, with some being in line with BL and others not. The BL paper was published in 2002, and its contribution was to explain forward prices in liberalized power markets with limited power storage. At that time, RES was limited in many markets,[6] and the liberalized markets were young. Since 2002, power markets have evolved with the entrance of new players, agents that learned about price dynamics, and the growth of RES. Koolen et al. (2022) observe price and quantity data from an experimental trading environment and show that firms having CES change their strategies in forward and spot markets. These energy market developments may explain why mixed results for the BL factors were observed.

In this paper, we propose a framework that extends BL in two ways. First, we introduce uncertain supply from RES, and RES producers that operate separately from conventional producers, and only trade in power spot markets. Second, we assume that agents have skewness preference in

[4]Reserve margin is the difference between installed production capacity minus demand (system load). It is a measure of market tightness and could also affect the market power.

[5]Market power is measured as the ratio of the power spot prices and the fundamental marginal cost estimate.

[6]Except for countries such as Norway with a large share of hydropower capacity.

their utility function besides the mean and variance. We motivate the mean-variance-skewness preferences assumption by observing that the empirical probability distribution function (PDF) of RES supply quantities, and therefore spot prices, is skewed.[7] The PDF of the RES supply is skewed because of capacity constraints. When the expected supply from RES is maximal (on sunny and windy days), actual quantities can only decrease, not increase, which produces a negatively skewed RES distribution. The opposite is true for days when we expect supply from RES to be low, and the shock will increase the quantity, resulting in a positively skewed RES distribution. Hence, depending on expected weather conditions, the PDF of RES is skewed. The summary statistics in Table 2 in the data section show that the empirical distribution function of demand is negatively skewed and that of RES supply is positively skewed. It also shows that the power spot prices are negatively skewed. We argue that agents in power markets with RES are aware of these skewness effects and account for skewness when making decisions.[8]

We first investigate the market agents' hedging behavior as a result of RES supply. The low marginal costs of RES crowd out CES in the spot market. Therefore, conventional producers with CES have a strong incentive to sell in the forward market to maintain market share, which exerts downward pressure on the forward price. The volatility of RES, on the other hand, incentivizes retailers to purchase forward, especially when renewable producers do not hedge. The skewness of RES affects in two ways. The first is a price effect, in which the agents would gain by selling more forward and buying spot if they expect the spot price to be possibly very low (negatively skewed), and vice versa. There is also a quantity effect due to uncertainty in the RES supply. Because an extremely low spot price is usually tied to a large RES supply, the quantity risk would amplify the price effect and can lead to extreme losses if such preferable states do not occur, hence creating an incentive to lower the forward position. The net change depends on which effect dominates.

We then derive a reduced-form expression for the equilibrium forward price that may help explain the mixed findings in the literature. We show that the forward price is a linear combination of the expected spot price, spot price variance, and skewness (as in the BL model), plus new components related to the covariance and coskewness between RES supply quantities and spot

[7] See Figures 1 and 2.

[8] In other words, when agents make hedging decisions, they not only think about smoothing profit volatility but also seek benefits (or avoid losses) from extreme prices. Skewness also matters because price jumps expose agents to asymmetric risks that they can't hedge away in spot markets.

prices. In addition to the missing components, when allowing for RES and skewness preferences, the signs for spot price variance and spot price skewness can be positive and negative (i.e., not strictly negative and positive, respectively, as predicted by BL). For instance, the role of spot price skewness depends on the market aggregate supply curve defined by marginal production costs. An increase in RES production lowers the demand that must be supplied by CES. If the installed CES capacity remains unchanged, production shifts to the flatter (less steep) part of the aggregate supply curve. In this case, a sudden increase in demand is less likely to trigger a spot price spike. Consequently, retailers face lower hedging pressure than under the BL setting, which reduces the positive effect of skewness on the forward price. As RES supply continues to grow, the spot price would often be low or even negative, and retailers would instead have incentives to hedge against this price dip, which might turn the sign negative.

Lastly, we empirically test our forward price model on data from the German power market. We find that the empirical results align with our theoretical predictions. We conclude that the supply risk coming from intermittent RES introduces interactions between spot prices and the uncertain supply from RES, which has an impact on the forward price. This effect is robust to both ex-ante and ex-post specifications. We propose that, besides the variance and skewness of spot prices from BL, covariance and coskewness between spot prices and RES supply should be considered as new components that explain the price of power forward contracts.

The rest of this paper is organized as follows. Section 2 builds the model. Section 3 describes the data and empirical methodology. Section 4 discusses empirical results. Section 5 concludes.

2 Model

We develop a two-period equilibrium model that extends the one of BL. Consider three types of agents: conventional power producers, renewable power producers, and retailers. They make decisions at two times. At time t_F, they decide quantities to buy or sell in the forward market. At time t_S, they decide quantities to buy or sell in the spot market (t_F is before t_S). We assume perfectly competitive markets, such that all agents are price takers. Let p_S be the spot price and p_F be the forward price for one unit of power. Furthermore, we assume the absence of government, arbitrageurs, and storage.

We begin by describing the agents and their profit functions. Then, we discuss demand and RES supply uncertainty and its impact on equilibrium power spot prices. We proceed with motivating the utility functions of the agents. After that, we analyze how they decide quantities to trade in forward markets. Finally, we derive the equilibrium forward price.

2.1 The agents

Consider three types of risk-averse agents. First, there is a representative conventional power producer, denoted by I, that converts fossil fuels into power. Second, there is a representative retailer, denoted by J, that sells power to consumers for an exogenous fixed price p_R. The retailer has no production capacity and purchases power in the forward and spot markets.[9] These two agents correspond to the agents in the BL model.[10] We extend their framework by introducing a representative renewable power producer, denoted by R, who uses RES such as wind and solar. We do not consider a producer that owns both CES and RES. Renewable producers often receive policy support and have a different risk profile compared with conventional producers, hence being operated separately. This extension of BL is not trivial, as the marginal costs and production flexibility of RES differ from those of CES. RES require no fuel, emit no CO2, and have no marginal production costs as a consequence. To the contrary, CES have positive marginal costs, including fuel costs and emission rights. Another difference is that supply from RES is intermittent as production quantities depend on varying weather conditions. Their production quantity cannot be flexibly adjusted, whereas CES can offer the flexibility to ramp up and down production quantities to match demand.[11]

[9]Hence, there is no vertical integration and all transactions occur through the forward and spot markets.

[10]Instead of modeling N_I conventional producers and N_J retailers explicitly, we use representative agents. This is without loss of generality under the assumption of homogeneous agents, since the aggregate behavior of many identical firms can be captured by a single representative. This aggregation absorbs N_I and N_J into the aggregate production cost function, aggregate demand function, and the risk-preference parameters, so that equilibrium outcomes depend only on aggregate capacity, demand, and risk attitudes. This ensures that the results are independent of the arbitrary number of identical participants, while retaining the essential mechanisms of the BL framework.

[11]The production of RES can be curtailed, which offers some ramp-down flexibility. In this paper, we don't consider the curtailment flexibility unless the renewable supply exceeds the total power demand.

2.2 The profit functions

We proceed with the profit functions of the representative agents. The representative conventional producer I sells a quantity Q_I^F in the forward market and Q_I^S in the spot market (negative values for Q_I^F and/or Q_I^S reflect purchases). Therefore, the conventional producer supplies a total quantity $Q_I = Q_I^F + Q_I^S$. The production cost function is

$$C(Q_I) = \frac{a}{c}(Q_I)^c, \qquad a > 0,\ c > 1, \tag{1}$$

where we ignore fixed costs that do not affect the results. The profit of the conventional producer is

$$\Pi_I = p_F Q_I^F + p_S Q_I^S - \frac{a}{c}(Q_I)^c. \tag{2}$$

The profit function of the representative renewable producer R is derived similarly. We assume that renewable producers only trade Q_R^S in the spot market and do not trade in the forward market, so $Q_R^F = 0$. This assumption is justified by the fact that renewable producers receive support for investing in renewable assets, and hence have less incentive to hedge in forward markets.[12] Furthermore, the renewable market is less centralized, with many relatively small wind and solar firms not having direct access to forward markets. The marginal production cost of RES is zero. As RES production is intermittent, the actual production quantity K is uncertain. The profit function of the renewable producer is

$$\Pi_R = p_S Q_R^S, \qquad Q_R^S = K. \tag{3}$$

Beyond the producers, the representative retailer J trades in the markets. The retailer takes a forward position Q_J^F, which is typically negative since the retailer is a net buyer of power, and purchases the difference between total power demand and forward position, $D + Q_J^F$, in the spot market. The profit function of the retailer is

$$\Pi_J = p_R D + p_F Q_J^F - p_S(D + Q_J^F). \tag{4}$$

[12] For instance, in Germany, almost all types of RES were eligible for a government-set feed-in tariff before 2014, after which the feed-in tariff was determined in an auction with a tender quantity cap. Awarded projects remain under price protection, while unawarded projects are more likely to be deferred to subsequent rounds rather than built without a price guarantee. See https://www.bundesnetzagentur.de/DE/Fachthemen/ElektrizitaetundGas/Ausschreibungen/start.html.

2.3 Demand and supply uncertainty

When agents choose their quantities to trade in the forward and spot markets, they face uncertainty that comes from two sources: consumer demand D (as in the BL model) and supply K from intermittent RES. We assume that both D and K are revealed at a time before the spot price is determined but after the time when agents make decisions in the forward markets (i.e., D and K are revealed somewhere between t_F and t_S). To simplify the analysis, we assume that $K \leq D$.[13] Both total demand and supply from RES are uncertain, i.e., $\mathrm{Var}(D) > 0$ and $\mathrm{Var}(K) > 0$.

We argue that a RES shock is not simply a negative demand shock. While a RES shock is influenced by weather, a demand shock can arise from many sources, with weather being only one of them. Table 2 supports this argument by showing the low correlation between D and K.

2.4 Spot market equilibrium

In the spot market, the renewable producer sells all power produced by RES, the retailer purchases power to satisfy demand, and the conventional producer sells power to maximize profit. As the market clears, the total production of the conventional producer equals the residual demand: $Q = D - K$. The market clearing spot price is determined by the CES producer maximizing its profit:

$$p_S = a(Q)^{1/x}, \qquad x = \frac{1}{c-1}. \tag{5}$$

We now analyze how the spot price changes when supply from RES is added to the market. Given demand, an increase in the RES power production decreases residual demand. Less quantity has to be produced by CES with positive marginal costs. As a consequence, the spot price decreases. The variance of spot price changes as well. The variance of residual demand is $\mathrm{Var}(Q) = \mathrm{Var}(D) + \mathrm{Var}(K) - 2 \times \mathrm{Cov}(D, K)$. When supply from RES increases, $\mathrm{Var}(Q)$ increases unless total demand and supply from RES co-move (i.e., $\mathrm{Cov}(D, K) > 0$). In the early stage of the energy transition, the focus is on adding RES to existing power systems and not on storage or demand

[13]Otherwise stated, we assume that any supply from RES that exceeds total demand is curtailed.

response.[14] Therefore, $\mathrm{Cov}(D, K)$ is close to zero and spot prices become more volatile.[15] Later, when more power storage facilities and demand response systems are added to energy systems, the covariance between D and K will increase as more (less) supply from renewables leads to higher (lower) demand response and/or power being injected in (withdrawn from) storage facilities. Then, the spot price volatility will decrease because of this positive covariance.

An increase in RES supply also affects the skewness of the spot price distribution. In a world with only CES, power is produced with a convex production cost function ($c > 2$). This convexity implies that a positive demand shock results in a larger absolute price change than a negative demand shock. The effect is most pronounced when the reserve margin is small. Then, positive demand shocks can result in extremely high prices. As a consequence, the distribution of spot prices is positively skewed (see BL, Barlow, 2002, and Borenstein, 2002). When RES supply increases, CES will produce power at a marginal production cost that comes from a region in the cost curve that is flatter.[16] Positive and negative demand shocks will then have a more equal impact on power prices than before, resulting in reduced positive skewness. See Figure 1.

2.5 Agent preferences

We have established how agents determine quantities to buy and sell in the spot market at time t_S. We now proceed with how they determine quantities of forward contracts at time t_F. As t_F is before t_S and before total demand D and supply from RES K are realized, agents make their decisions under uncertainty about demand, supply, spot prices, and thus their profits. Therefore, we must specify the agents' utility from uncertain future profits. While BL assume that all agents share a mean–variance utility function, we instead assume they follow a mean–variance–skewness utility function:

$$U(\Pi) = \mathbb{E}(\Pi) - \frac{1}{2\gamma}\ \mathrm{Var}(\Pi) + \frac{1}{3\phi}\ \mathrm{Skew}(\Pi), \tag{6}$$

[14]In 2022, wind and solar accounted for about 22% of power production in Europe, while contributions from battery storage and demand response remained negligible.

[15]When we refer to the covariance between demand and RES production, we mean the comovement of their unexpected shocks. The well-known pattern that solar generation co-moves with power demand in the afternoons is predictable and therefore not considered uncertain. Since power demand is highly inelastic and RES production is intermittent and largely uncontrollable, the covariance is typically very small.

[16]To be precise, we consider the medium-run scenario where CES capacity remains fixed. In the long run, CES capacity would also adjust in response to larger RES capacity.

where $\gamma > 0$ is the risk tolerance parameter and $\phi > 0$ is the skewness tolerance parameter. A smaller γ implies a larger degree of risk aversion. A smaller ϕ represents a more pronounced skewness preference. Consistent with the literature, risk-averse agents dislike profit variance (hence the negative sign for Var) but prefer positive skewness in the profit distribution (hence the positive sign for Skew). A positive ϕ reflects agents' preference for taking risks that involve low-probability, high-return events, and a willingness to pay insurance premiums to avoid rare but extreme losses.

The inclusion of skewness in the utility function is relevant only if the profit distribution is asymmetric, i.e., when $\text{Skew}(\Pi) = \mathbb{E}\left[(\Pi - \mathbb{E}[\Pi])^3\right] \neq 0$. This condition is easily satisfied in power markets because the conventional producer's cost function, and hence its profit function, is non-linear in spot prices. This also applies to retailers, who face both price and quantity risk, resulting in asymmetric profit distributions. Furthermore, electricity prices are highly asymmetric: price spikes have often been observed in the past, and the skewness of the RES supply distribution propagates into the distribution of spot prices and consequently into profits.[17] This skewness is not well captured by variance alone. Hence, we want to build a model in which agents care not only about the average and volatility of returns, but also about the likelihood of extreme outcomes. Skewness preferences capture this asymmetry.

In the broader asset pricing literature, skewness preference has strong theoretical and empirical foundations. Theoretically, this cubic utility function is justified by preferences for higher-order moments from a risk-averse agent (Kraus & Litzenberger, 1976).[18] Harvey and Siddique (2000) show that conditional skewness can explain the cross-sectional differences in stock returns, with systematic skewness priced in the market and requiring a risk premium. Also, Boyer et al. (2010) find evidence that investors are willing to pay a premium for assets with high expected idiosyncratic

[17]To see this, consider three scenarios. In Scenario 1, weather conditions for RES are moderate, so both RES and CES are used to meet demand. Changes in RES supply can be offset by CES adjusting production, with limited impact on spot prices. In Scenario 2, weather conditions are bad for RES (no wind, thick clouds), so the expected RES supply is close to zero. Any positive RES shock reduces residual demand, leading to a left-skewed distribution of spot prices. As a result, the profit distribution of conventional producers becomes less positively skewed (or more negatively skewed), while that of retailers becomes less negatively skewed (or more positively skewed). In Scenario 3, weather conditions are ideal for RES (strong wind, clear skies), so RES supply is high. Negative RES shocks increase residual demand, generating a right-skewed spot price distribution. Consequently, the profit distribution of conventional producers becomes more positively skewed (or less negatively skewed), while that of retailers becomes more negatively skewed (or less positively skewed).

[18]For instance, if we assume a CARA utility form $U(\Pi) = -e^{-A\Pi}$, a third-order Taylor expansion and certainty equivalence imply a mean-variance-skewness utility function such that $\gamma = \frac{1}{A}$ and $\phi = \frac{2}{A^2}$.

skewness.[19]

To sum, profit distributions in power markets are inherently asymmetric, and the transmission of skewed RES supply into spot prices could change the asymmetry of profit and this could be perceived by market participants as in other financial and commodity markets. Therefore, we think that mean-variance preferences (as BL assume) are too restrictive to model power markets with RES, and we propose to include skewness in the agent preferences in equation 6.

2.6 Forward market equilibrium

In this section, we discuss the equilibrium conditions in the power forward market. We begin with the optimal forward positions Q_I^F (for the conventional producer) and Q_J^F (for the retailer) that maximize their expected utility.[20] Then, we derive the equilibrium forward price and its reduced-form representation.

2.6.1 The optimal forward positions

The utility-maximizing forward positions of the representative agents are implicitly given by

$$Q_{I(J)}^{F^*} = \underbrace{\gamma \cdot \frac{p_F - \mathbb{E}(p_S)}{\mathrm{Var}(p_S)}}_{\text{basis term}} + \underbrace{\frac{\mathrm{Cov}(\rho_{I(J)}, p_S)}{\mathrm{Var}(p_S)}}_{\text{hedging term}} + \underbrace{\frac{\gamma}{3\phi} \cdot \frac{\partial\, \mathrm{Skew}(\Pi_{I(J)})/\partial Q_{I(J)}^F}{\mathrm{Var}(p_S)}}_{\text{skewness term}}, \tag{7}$$

where

$$\rho_I = p_S Q_I - \frac{a}{c}(Q_I)^c, \qquad \rho_J = (p_R - p_S)D,$$

are the *but-for-hedging* profits, i.e. profits earned with zero forward position. Throughout, the notation $I(J)$ indicates that the expression applies separately to the conventional producer I and

[19]Skewness preference is also referred to as prudence in saving decisions (Kimball, 1990). In the prospect theory model, this assumption is rationalized as loss aversion, describing people's tendency to prefer avoiding loss over gaining equivalent gains (Tversky & Kahneman, 1986, 1992). While these models emphasize downside risk, we also allow for a lottery preference, i.e., the desire for extremely high payoffs. See (Barberis & Huang, 2008).

[20]To find a solution, we need to ensure that the mean-variance-skewness utility function 6 is concave.[21] Therefore, we restrict the utility maximising forward position $Q^F \in [\underline{q}, \bar{q}]$ such that $\frac{\partial U^2(\Pi(Q^F))}{\partial (Q^F)^2} \leq 0$. This has two consequences. First, the agents are not allowed to buy or sell an infinite number of forward contracts. Second, aversion to profit volatility takes a larger weight than preference for skewness, implying that the optimal forward position with mean-variance-skewness preferences should not be too far from the optimal position when only mean-variance preferences are assumed.

the retailer J.

The optimal forward positions in equation 7 depend on three terms: a basis term, a hedging term, and a skewness term. The basis term captures agents' incentive to exploit the forward–spot price difference. When the forward price exceeds the expected spot price, agents sell more in the forward market. The basis term is proportional to the risk tolerance parameter γ (recall that a smaller γ indicates higher risk aversion). More risk-averse agents sell fewer forwards. The hedging term is the normalized sensitivity of an agent's profit to changes in the spot price when unhedged (i.e. when the forward position is zero). The profit of the conventional producer increases when spot prices rise, which makes the hedging term positive. The producer sells more forward contracts when profits are more sensitive to spot price changes. In contrast, the retailer faces losses when spot prices rise, so its hedging term is negative. To manage profit risks, the retailer purchases more forward contracts when profits are more sensitive to spot price changes.

Although the basis and hedging terms have the same form as in BL, the optimal forward positions are different, as RES supply affects the spot price level and uncertainty. When we allow for skewness preference, the skewness term emerges in equation 7 that reflects an agent's incentive to take advantage of the asymmetry of the profit distribution function. To see how agents benefit from skewness, first assume that they hold the optimal position as in the BL model, i.e., the forward position under mean-variance preferences. It must hold that $\frac{\partial[\mathbb{E}(\Pi)-\frac{1}{2\gamma}\operatorname{Var}(\Pi)]}{\partial Q^F} = 0$. But if $\partial \operatorname{Skew}(\Pi)/\partial Q^F \neq 0$, a small change in the forward position increases an agent's utility. The agent adjusts its optimal forward position until the marginal loss from the basis and hedging terms equals the marginal benefit from the skewness term.[22] The extent to which an agent adjusts its forward position depends on the risk and skewness tolerance parameters γ and ϕ. If agents are very risk-averse ($\gamma \approx 0$), the skewness term becomes negligible. The more an agent values extreme profits (a smaller ϕ), the more important the skewness term becomes in determining the forward position. To determine the direction of forward position change, we need to know the sign of $\frac{\partial \operatorname{Skew}(\Pi)}{\partial Q^F}$. Taking the derivative of profit skewness w.r.t the forward position gives

$$\frac{\partial \operatorname{Skew}(\Pi)}{\partial Q^F} = -3\operatorname{Skew}(p_S)(Q^F)^2 + 6\operatorname{Coskew}(\rho, p_S^2)(Q^F) - 3\operatorname{Coskew}(\rho^2, p_S), \tag{8}$$

[22]When $Q^F = Q^{F^*}$, $-\frac{\partial[\mathbb{E}(\Pi)-\frac{1}{2\gamma}\operatorname{Var}(\Pi)]}{\partial Q^F} = \frac{\partial \frac{1}{3\phi}\operatorname{Skew}(\Pi)}{\partial Q^F}$.

where $\text{Coskew}(X, Y, Z)$ denotes the third mixed central moment $\mathbb{E}\{[X - \mathbb{E}(X)][Y - \mathbb{E}(Y)][Z - \mathbb{E}(Z)]\}$. For compactness, $\text{Coskew}(X^2, Z)$ denotes $\text{Coskew}(X, X, Z)$. Thus,

$$\text{Coskew}(\rho^2, p_S) = \mathbb{E}\{[\rho - \mathbb{E}(\rho)]^2[p_S - \mathbb{E}(p_S)]\},$$
$$\text{Coskew}(\rho, p_S^2) = \mathbb{E}\{[\rho - \mathbb{E}(\rho)][p_S - \mathbb{E}(p_S)]^2\}.$$

The RHS of equation 8 has a quadratic form in Q^F. Therefore, the profit skewness does not necessarily change monotonically with the forward position Q^F. To see this, consider the case that $\text{Skew}(p_S) < 0$.[23] When there is no quantity risk, meaning that the *but-for-hedging* profit is linear in the spot price, the minimum of equation 8 is zero. As a consequence, $\frac{\partial\,\text{Skew}(\Pi)}{\partial Q^F} \geq 0$, i.e., an increase in the forward position increases profit skewness. The intuition is straightforward: a negatively skewed spot price distribution implies a small probability of extremely low spot prices, allowing the agent to make a large profit by selling in the forward market and buying in the spot market. We refer to this as the price effect.

However, both conventional producers and retailers face quantity risk. For retailers, the quantity risk is unexpected surges or drops in consumers' power demand. Conventional producers face total demand risk and manage sudden changes in supply from RES, hence the residual demand risk. If residual demand is positively skewed, conventional producers may face states with both high spot prices and high demand, which amplify the extreme gain when they choose not to sell much forward, but also imply low profits if such states do not occur. Hence, the producer could benefit by increasing their forward sales. The analysis for negatively-skewed residual demand is similar. For retailers, large forward purchases can lead to substantial losses if unexpectedly high demand does not materialize, creating an incentive to reduce forward exposure, and vice versa. We refer to this as the quantity effect. The net effect depends on which effect is larger.

Proposition 1 *Skewness preference affects optimal forward positions through both price and quantity effects.*

1. ***Price effect:***

    - $\text{Skew}(p_S) < 0$*: both conventional producers and retailers sell more forwards.*

[23]The analysis of the case that spot price is positively skewed, $\text{Skew}(p_S) > 0$, is similar.

- $\text{Skew}(p_S) > 0$*: both conventional producers and retailers sell fewer forwards.*
- $\text{Skew}(p_S) = 0$*: no effect.*

2. ***Quantity effect:***
   - $\text{Skew}(Q) < 0$*: conventional producers sell fewer forwards.*
   - $\text{Skew}(Q) > 0$*: conventional producers sell more forwards.*
   - $\text{Skew}(Q) = 0$*: no effect on conventional producers.*
   - $\text{Skew}(D) < 0$*: retailers sell fewer forwards.*
   - $\text{Skew}(D) > 0$*: retailers sell more forwards.*
   - $\text{Skew}(D) = 0$*: no effect on retailers.*

2.6.2 The equilibrium forward price

As shown in Appendix C, the equilibrium forward price is determined by the market clearing condition: $Q_I^F + Q_J^F = 0$. We first summarize the results below.[24]

Proposition 2 *Define*

$$\tau_{I(J)} = \frac{\text{Coskew}(\rho_{I(J)}, p_S^2)}{\text{Var}(p_S)} - \frac{\phi}{2\gamma}, \tag{9}$$

$$p_{I(J)} = \mathbb{E}(p_S) - \frac{1}{\gamma}\text{Cov}(\rho_{I(J)}, p_S) + \frac{1}{\phi}\text{Coskew}(\rho_{I(J)}^2, p_S), \tag{10}$$

$$w_{I(J)} = \frac{\tau_{J(I)}}{\tau_I + \tau_J}. \tag{11}$$

Only if $\tau_I + \tau_J < 0$*, the equilibrium forward price exists and is given by:*

$$p_F = w_I p_I + w_J p_J + \frac{\phi}{4}\frac{\text{Skew}(p_S)}{\text{Var}(p_S)^2}\frac{(p_I - p_J)^2}{(\tau_I + \tau_J)^2}. \tag{12}$$

τ_I and τ_J represent composite risk measures for the conventional producer and retailer, respectively. It has two components: $\frac{\text{Coskew}(\rho_{I(J)}, p_S^2)}{\text{Var}(p_S)}$, which measures the *but-for-hedging* profit sensitivity per unit of spot-price variance (i.e., exposure to tail risk normalized by volatility), and $\frac{\phi}{2\gamma}$,

[24] In the absence of skewness preference, $\phi = \infty$, the forward price reduces to the simple average $p_F = \frac{1}{2}(p_I + p_J)$.

the ratio of skewness preference versus variance aversion. It is positive when the skewness tolerance parameter ϕ is sufficiently small—implying a strong preference for skewness—or when *but-for-hedging* profits largely increase with spot price volatility. Intuitively, a larger τ indicates a weaker incentive to hedge. Their sum, $\tau_I + \tau_J$, captures the industry risk preferences. When the sum is positive, at least one agent lacks incentive to hedge, so there is no matched forward position. Hence, a forward equilibrium exists only if aggregate variance aversion dominates skewness preference, i.e., when $\tau_I + \tau_J < 0$. The equilibrium forward positions are:

$$Q_I^F = \frac{\phi}{2\,\mathrm{Var}(p_S)} \frac{p_I - p_J}{\tau_I + \tau_J}, \tag{13}$$

$$Q_J^F = -Q_I^F. \tag{14}$$

Normally, the conventional producer sells forwards while the retailer buys forwards, implying $p_I < p_J$.[25] When both the producer and retailer exhibit variance aversion dominating skewness preference, i.e., $\tau_I < 0$ and $\tau_J < 0$, p_I and p_J can be interpreted as the *reservation prices* at which the conventional producer and retailer, respectively, optimally choose a zero forward position. Since the optimal forward position is increasing in the forward price, the equilibrium forward price must lie strictly between p_I and p_J, $p_I < p_F < p_J$. The weight $w_{I(J)}$ shows the relative risk measure, and the less risk-sensitive agent should have a larger weight. Intuitively, if an agent is risk-neutral, they trade on the forward market for arbitrage reasons. Hence, when the forward price is far from their reservation price, they will continue trading until all arbitrage opportunities are eliminated.

Now consider the case where either $\tau_I \geq 0$ or $\tau_J \geq 0$, but $\tau_I + \tau_J < 0$. In this situation, even though an equilibrium may still exist in theory, the concavity of the utility function, which ensures a well-behaved optimum, becomes weak. This local maximum is typically small and fragile: even minor deviations from the equilibrium position increase utility, making the optimum unstable. For example, if the producer exhibits a strong skewness preference and anticipates a positively skewed spot price distribution, it will prefer to buy forward and liquidate in the spot market, while the retailer also wants to buy forwards for hedging. Since no counterparty is willing to take the short

[25]Reverse trading is possible, with the producer buying forwards and the retailer selling forwards, so $p_I > p_J$. For clarity, we focus on the typical case.

side, the forward market cannot clear.[26]

2.6.3 Reduced-form representation of the forward premium

By the equilibrium spot price given in equation 5 and the spot market clearing condition, the *but-for-hedging* profits, ρ_I and ρ_J, can be expressed as:

$$\rho_I = \frac{1}{(x+1)a^x} p_S^{x+1},$$
$$\rho_J = (p_R - p_S)K + (p_R - p_S)\frac{p_S^x}{a^x}.$$

To compare with BL and derive a theoretical prediction testable by the central moments of wholesale spot prices and the mixed moments between renewable production and wholesale spot prices, we assume that the normalized risk measures τ and the weights w remain constant under changes of spot price variance. Then, we approximate p^x and p^{x+1} using second-order Taylor series expansions shown in Appendix D and rewrite the equilibrium forward premium as

$$\begin{aligned} p_F - \mathbb{E}(p_S) = {} & \alpha \operatorname{Var}(p_S) + \beta \operatorname{Skew}(p_S) \\ & + \theta \operatorname{Cov}(K, p_S) + \xi \operatorname{Coskew}(K^2, p_S) + \eta \operatorname{Coskew}(K, p_S^2). \end{aligned} \tag{15}$$

[26]In other words, when agents exhibit strong skewness preference, their distinct roles as producer and retailer become irrelevant, and they treat power forwards as purely financial assets rather than instruments to manage trading risk.

where

$$
\begin{aligned}
\alpha &= -w_I \frac{[\mathbb{E}(p_S)]^x}{\gamma a^x} - w_J \left[\frac{[\mathbb{E}(p_S)]^{x-1}}{\gamma a^x} \big(x p_R - (x+1)\mathbb{E}(p_S) \big) - \mathbb{E}(K) \right], \\
\beta &= w_I \left[\frac{(x-1)^2 [\mathbb{E}(p_S)]^{2x}}{\phi a^{2x}} - \frac{x [\mathbb{E}(p_S)]^{x-1}}{2\gamma a^x} \right] \\
&\quad + w_J \left[\frac{\psi^2}{\phi a^{2x}} - \frac{x [\mathbb{E}(p_S)]^{x-2}}{2\gamma a^x} \big((x-1) p_R - (x+1)\mathbb{E}(p_S) \big) \right] \\
&\quad + \frac{\phi}{4\gamma(\tau_I + \tau_J)^2} \left[\frac{[\mathbb{E}(p_S)]^{x-1}}{a^x} \big(x p_R - (x+2)\mathbb{E}(p_S) \big) - \mathbb{E}(K) \right]^2, \\
\theta &= -\frac{w_J}{\gamma} \big(p_R - \mathbb{E}(p_S) \big), \\
\xi &= \frac{w_J}{\phi} p_R^2, \\
\eta &= w_J \left[\frac{2\psi p_R}{a^x \phi} + \frac{1}{\gamma} \right], \\
\psi &= [\mathbb{E}(p_S)]^{x-1} \left[x(2-x) p_R + (x+1)(x-1)\mathbb{E}(p_S) \right].
\end{aligned}
$$

The first observation is that the variance and skewness of spot prices are insufficient to explain the forward premium when uncertainty comes from demand D and RES supply K. When demand and supply shocks are not perfectly correlated, as argued in section 2.4 and empirically shown in Table 2, K explicitly appears in the equilibrium equation 15.

Another observation is that, although the variance and skewness factors were found by BL, their coefficients change with the inclusion of RES and skewness preference. BL predict $\alpha < 0$. This can be explained as follows. The conventional producers are willing to sell forward contracts for hedging reasons. On the other side, retailers' aversion to price risk drives them to purchase forward contracts. When conventional producers and retailers have the same risk preference, these two hedging pressures offset, and there is no bias on the forward price. However, since high spot prices correlate with high power demand, the additional profit margins produce a counteracting incentive for retailers to sell forward contracts. This leads to a net selling of forward contracts, thereby exerting downward pressure on the forward premium ($\alpha < 0$). However, the expression for α in equation 15 shows that the BL rationale may not hold if conventional producers face greater quantity risk than retailers ($\tau_I < \tau_J$), giving them an incentive to sell fewer forwards and thereby driving up the forward premium, or if the expected RES supply K is sufficiently large, incentivizing retailers to buy more forwards to hedge the uncertainty from RES, which also increases the forward premium. As a consequence, we cannot determine the sign of α when we allow for supply from RES and/or skewness preferences.

The determinants of the parameter β change as well. Note that β depends on both the risk tolerance γ and skewness tolerance ϕ. When risk tolerance is sufficiently large, β is positive and driven by skewness preference. Otherwise, there is a trade-off between the incentive to hedge price variance and to profit from price skewness. BL predict $\beta > 0$ because retailers are exposed to more profit risk from unexpected demand surges (negative demand shock and negative price shock) than conventional producers that benefit from high spot prices. Therefore, retailers have a larger incentive to buy forward contracts, which increases forward prices. However, as RES supply grows, reliance on CES decreases. The conventional producers operate in a flatter region of their marginal cost function, leading to less frequent high spot prices. Hence, there is less demand for hedging the price variance and the forward price decreases. However, when the share of RES in power production becomes sufficiently large, the price variance increases again, and retailers will hedge for this risk. Because the variance of spot price is now related to a more negatively skewed spot price distribution, the hedging incentive in β has a negative rather than positive attribute as in BL. When the risk tolerance is smaller than skewness tolerance, β could turn negative and drive up the forward price. In general, the sign of β cannot be predicted ex-ante. The complex composition of the parameters α and β may explain the mixed empirical findings from the BL model in the literature.

Our next contribution is that we suggest three components that explain the forward premium beyond spot price variance and skewness. The components capture the relationship between spot prices and RES supply. The first component is the covariance between RES supply and spot prices, $\mathrm{Cov}(K, p_S)$, which is negative since more RES supply reduces spot price. This component was found by Schwenen and Neuhoff (2024) as well. They propose (and find empirical support) that the forward premium increases when $\mathrm{Cov}(K, p_S)$ decreases. That is, they expect $\theta < 0$, as we do. More RES supply implies that CES produces less quantity, and the total supply of forward contracts decreases as a consequence. The forward premium then increases. Note that we find this covariance factor under different assumptions for the market. Schwenen and Neuhoff (2024) do not model demand uncertainty. Instead, they model spot price uncertainty without specifying its source. Furthermore, their agents do not have skewness preferences. Therefore, our support of the covariance factor in explaining the forward premium is more robust.

The other two components in equation 15 are the co-skewness factors that result from retailers' skewness preference. The first co-skewness, $\text{Coskew}(K^2, p_S)$, reflects information about how uncertainty of the supply from RES (as expressed by K^2) affects changes in spot prices. RES supply uncertainty can manifest in two ways. When the distribution of RES supply has a right tail, there is a low probability of a large RES supply increase and an extremely low spot price. Because retailers prefer low spot prices, they sell in the forward market, exerting downward pressure on the forward price and premium. To the contrary, if the RES supply is negatively skewed, there is a likelihood of an extremely high spot price, and retailers purchase more forwards, thereby driving up the forward price. Therefore, the model predicts that $\xi > 0$.

The second coskewness term captures the relationship between spot price uncertainty (p_S^2) and RES supply. The sign of its coefficient η cannot be determined. There are two key effects. First, when RES supply grows, spot price variance increases, and retailers purchase more forward contracts due to their risk aversion, so the forward price increases to clear the market. Second, although spot price variance increases, its impact depends on the skewness of the spot price distribution. If the increased variance is associated with a decline (rise) in spot prices that benefits (harms) retailers, they will have less (more) incentive to buy forward, leading to a decrease (increase) in the forward price. This effect resembles the role of $\text{Coskew}(K^2, p_S)$. The sign of η depends on the relative strength of these two forces: $\eta > 0$ if risk aversion (variance term) dominates, and $\eta < 0$ if skewness preference plays a larger role.

It is straightforward to verify that, without uncertain RES supply K and skewness preferences, the equilibrium forward premium 15 reduces to the BL model. When skewness preferences are introduced, the forward premium is still only driven by the variance and skewness of spot prices. However, the weights $w_I, w_J \neq \frac{1}{2}$, and the signs of α and β become indeterminate. This helps explain the mixed empirical findings in the literature, even when the share of intermittent renewables is small. If skewness preferences are abstracted from and only uncertain RES supply is introduced, two additional risk components, $\text{Cov}(K, p_S)$ and $\text{Coskew}(K, p_S^2)$, emerge, and the sign predictions are clear: $\theta < 0$, and $\eta > 0$.

It is important to note that all parameters in equation 15 depend on time-varying variables such as expected spot prices and supply from RES. Furthermore, we cannot expect that the level of other parameters, such as the agent's risk and skewness tolerance, the CES production costs, and the retail price, will remain constant over time. Hence, we expect the magnitude of the effects of the risk measures to vary over time, while the predicted signs follow the same intuition. We summarize the main implications of the parameters in proposition 3.

Proposition 3 *When both demand and supply from RES are uncertain and market participants have mean-variance-skewness preferences, the forward premium of a power forward contract is a linear combination of spot price variance* $\mathrm{Var}(p_S)$*, spot price skewness* $\mathrm{Skew}(p_S)$*, covariance* $\mathrm{Cov}(K, p_S)$*, co-skewness* $\mathrm{Coskew}(K^2, p_S)$*, and co-skewness* $\mathrm{Coskew}(K, p_S^2)$*. The forward premium is negatively correlated with the covariance and positively correlated with the co-skewness between the spot price and the variance of renewable production,* $\theta < 0, \xi > 0$*. The signs of parameters* α, β*, and* η *are indefinite.*

3 Empirical methodology and data

We proceed with applying the forward premium equation 15 to market data, where the premium is defined as $p_F - \mathbb{E}(p_S)$. This is the ex-ante forward premium relating the forward price to the expected spot price during a future delivery period, while the expected spot price is not directly observed in data. Hirshleifer (1989) distinguishes between ex-ante and ex-post forward premiums. Ex-ante forward premiums provide information about whether the related risks are priced. Ex-post forward premiums, defined as the forward price minus the realized average spot price, provide insight into the realized premiums earned for taking the risks. We test the model both ex-ante and ex-post.

We begin with how we test our model ex-post. The regression equation is

$$\begin{aligned} p_{F_{t,T}} - p_{S_T} = {} & \alpha \ \mathrm{Var}(p_{S_T}) + \beta \ \mathrm{Skew}(p_{S_T}) \\ & + \theta \ \mathrm{Cov}(K_T, p_{S_T}) + \xi \ \mathrm{Coskew}(K_T^2, p_{S_T}) \\ & + X_{t,T} + \mathrm{MFE}_T + \epsilon_t . \end{aligned} \tag{16}$$

$p_{F_{t,T}}$ is the price of the futures contract for delivery in month T, observed on the last trading day of the preceding month ($t \in T-1$).[27] The realized spot price p_{S_T} is the monthly average spot price in T. We have hourly day-ahead spot prices, and we distinguish between base and peak load futures contracts. We compute the daily average base and peak prices first. Let $p_{S_{t,h}}$ be the spot price for delivery of one MWh during hour h on day t.[28] Define B_t and P_t as the sets of base hours and peak hours. n_{B_t} is the number of baseload hours and n_{P_t} is the number of peak load hours on day t.[29] n_T is the number of relevant days in month T.[30] The daily averages are

$$p_{S_{B,t}} = \frac{1}{n_{B_t}} \sum_{h \in B_t} p_{S_{t,h}}, \qquad p_{S_{P,t}} = \frac{1}{n_{P_t}} \sum_{h \in P_t} p_{S_{t,h}}.$$

The monthly averages are

$$p_{S_T} = \frac{1}{n_T} \sum_{t \in T} p_{S_{B,t}} \quad \text{(baseload)}, \qquad p_{S_T} = \frac{1}{n_T} \sum_{t \in T} p_{S_{P,t}} \quad \text{(peak load)}.$$

We follow the same logic to compute the other variables in the regression equation. For example, $\mathrm{Var}(p_{S_T})$ is the variance calculated as deviations of daily averages from the monthly averages (either for base or peak load prices, whichever applies).

Note that we have not included $\mathrm{Coskew}(K, p_S^2)$ from equation 15 in the regression equation, as we observed that both co-skewness variables are highly correlated in our sample, leading to multicollinearity issues. We proceed with $\mathrm{Coskew}(K^2, p_S)$. The reason is that we predict a positive sign for this variable, whereas the sign for the other is undetermined. Furthermore, we think that $\mathrm{Coskew}(K^2, p_S)$ has a clearer economic interpretation as it is directly derived from the skewness preferences and relates to the uncertainty of supply from RES and the spot price.[31]

Last, equation 16 includes $X_{t,T}$ and MFE_T. $X_{t,T}$ represents control variables such as conventional fuel prices and prices of carbon emission allowance. MFE_T represents month dummy variables

[27] We use the forward price from the last trading day, as it reflects the most accurate information about the delivery month. We also tested using (i) the average futures price over all trading days in $T-1$ and (ii) the average over the last trading week. The results are robust.

[28] As we use day-ahead prices for spot prices, the spot prices are actually observed one day before delivery.

[29] B_t contains 24 hours for most days except for days when time is adjusted for daylight saving time.

[30] Note that the number of days for peak load is less than for base load as peak load only applies to working days.

[31] The correlation between $\mathrm{Coskew}(K^2, p_S)$ and $\mathrm{Coskew}(K, p_S^2)$ is close to -1, so including both variables merely reallocates the estimates on $\mathrm{Coskew}(K^2, p_S)$ and $\mathrm{Coskew}(K, p_S^2)$. The covariance term $\mathrm{Cov}(K, p_S)$ captures the relationship between the level of RES supply and the spot price. The coskewness term $\mathrm{Coskew}(K^2, p_S)$ extends this by adding the effect of uncertainty in RES supply on spot prices. By contrast, $\mathrm{Coskew}(K, p_S^2)$ measures how the level of RES supply interacts with uncertainty in the spot price itself.

that absorb predictable seasonal patterns. We applied the same methodology for calculating the monthly averages for demand D, residual demand Q, and power quantities supplied by renewables K.

When we test our model ex-ante, the regression equation is as follows:

$$\begin{aligned} p_{F_{t,T}} - E_t(p_{S_T}) = {} & \alpha \operatorname{Var}_t(p_{S_T}) + \beta \operatorname{Skew}_t(p_{S_T}) \\ & + \theta \operatorname{Cov}_t(K_T, p_{S_T}) + \xi \operatorname{Coskew}_t(K_T^2, p_{S_T}) + \\ & + X_{t,T} + \mathrm{MFE}_{t,T} + \epsilon_t. \end{aligned} \tag{17}$$

The forward premium is $p_{F_{t,T}} - E_t(p_{S_T})$, meaning the difference between the forward price observed for delivery during month T observed at time t and the average spot price for month T as expected at time t. There is no general way to measure the expected spot price for a future time period. Agents in the energy market use different proprietary models to make expectations. We choose a very simple approach. We measure the expected spot price $E_t(p_{S_T})$ as the (rolling) average spot price observed over the previous 30 days before t. The drawback of this backward looking method is that it ignores any ”predictable” changes in the spot prices (for instance, when T is a month with a normally higher demand than for month $T-1$ one would expect a higher spot price during month T). To account for such effects, we add month dummy variables (the month fixed effects $\mathrm{MFE}_{t,T}$). We apply the same backward looking method to measure the risk factors variance, skewness, covariance, and co-skewness (hence the time subscripts $_t$ for those variables as the rolling estimates vary over different t's. The advantage of the rolling estimates in the ex-ante analysis is that we can use observations from every day in the month prior to the delivery period and don't have to rely on the last trading day only as we do for the ex-post analysis. Next, we describe the data that we use.

3.1 Data

Our sample consists of data from the German power markets (EPEX for spot contracts and EEX for futures contracts) provided by ENTSO-E, SMARD, and Refinitiv.[32] The sample covers the period January 2015 through April 2021, and consists of the following variables:[33]

- day-ahead hourly power prices (we refer to them as spot prices),
- day-ahead hourly supply quantities from wind and solar RES,
- day-ahead hourly power demand,
- daily one-month-ahead power baseload and peak load futures prices,
- daily one-month-ahead coal (API2 Coal, \$/MT) futures prices,
- daily one-month-ahead natural gas (RFV Natural Gas, €/MWh) futures prices,
- daily carbon emission allowances futures prices (EUA, €/ton).

We use estimated marginal generation costs for natural gas and coal-fired power plants, expressed in €/MWh, as control variables ($X_{t,T}$ in equations 16 and 17). These costs are the sum of fuel costs and carbon costs:

$$C_t^{\text{gas}} = \frac{P_t^{\text{gas}}}{\eta_{\text{gas}}} + e_{\text{gas}} P_t^{\text{CO2}},$$
$$C_t^{\text{coal}} = \frac{P_t^{\text{coal}}}{\eta_{\text{coal}}} + e_{\text{coal}} P_t^{\text{CO2}}.$$

Here, P_t^{gas} and P_t^{coal} are fuel prices expressed in €/MWh, and P_t^{CO2} is the carbon allowance price in € per ton of CO_2. Coal prices quoted in US dollars per metric ton are converted to equivalent prices in €/MWh. We assume thermal efficiencies of $\eta_{\text{gas}} = 0.5$ for gas-fired plants and $\eta_{\text{coal}} = 0.4$ for coal-fired plants, and emission factors of $e_{\text{gas}} = 0.35$ tons of CO_2/MWh for gas and $e_{\text{coal}} = 0.95$ tons

[32] The power day-ahead prices are determined per bidding zone. The German bidding zone covered Germany, Luxembourg, and Austria until October 2018. After that, it covered Germany and Luxembourg. As the share of German power in this bidding zone is much higher than the share of Luxembourg power, we refer to this as Germany.

[33] In total, the dataset covers 76 months, including 55488 hours of power spot prices. There are 1652 weekdays with available baseload and peak-load futures prices, and 1600 weekdays (holiday excluded) with gas, coal, and carbon emission allowance futures prices. The number of observations varies slightly due to missing values for some variables.

of CO_2/MWh for coal. These values are standard approximations for European thermal generation costs (Fabra & Reguant, 2014; Fleten et al., 2015).

3.2 Summary statistics

Panel A of Table 2 provides descriptive statistics of the data described in section 3.1. Panel B and Panel C of Table 2 report risk measures calculated based on the method described in section 3, for baseload and peak load, respectively. Focusing on baseload, we observe that the average hourly demand was 59.75 GWh. Obviously, during peak hours, the average demand is higher than during a base load hour. Demand D is negatively skewed. Supply from RES K is positively skewed, and is higher during peak hours as both solar and wind produce during peak hours, whereas solar cannot produce during nighttime (off-peak). Residual demand is more variable than demand and supply from RES. This can be explained by the fact that the correlation between demand and RES supply is weak in our sample.[34] As a consequence, $\mathrm{Var}(Q) \approx \mathrm{Var}(D) + \mathrm{Var}(K)$.

Spot prices are negatively skewed. Also, we find that the covariance between renewable supply and spot prices is negative in both base and peak hours. This is consistent with equation 5 that the spot price depends on residual demand $(D - K)$. Increases in RES supply reduce the residual demand to be supplied by conventional producers, which lowers the spot price. The covariance is more negative during peak hours, as RES shocks have more impact at times with high demand. When demand is high, a negative supply shock will lead to a higher marginal price. The coskewness terms $\mathrm{Coskew}(K^2, p)$ are also negative. This implies that increased uncertainty in supply from RES yields lower spot prices. We explain this from a positively skewed supply from renewables, driving negatively skewed spot prices. Finally, forward premiums are on average positive and slightly higher for delivery during peak hours.

4 Empirical results

In this section, we test our hypotheses by the regression models 16 and 17. We start with discussing the results for the ex-ante forward premiums. Table 3 shows the results for the base load futures

[34] The correlation coefficient between demand and renewable production is almost zero during base load hours and 0.07 during peak hours.

Table 2: **Summary Statistics**

Panel A reports descriptive statistics of day-ahead hourly power prices (spot prices), day-ahead hourly supply quantities from wind and solar RES, day-ahead hourly power demand, one-month-ahead power baseload and peak load futures prices, one-month-ahead coal and natural gas futures prices, and carbon emission allowances futures prices. Ex-post baseload and peak load forward premiums are calculated by the difference between the observed forward price and the average spot price. Panel B and Panel C report ex-post risk measures calculated following section 3, for baseload and peak load, respectively. The coskewness is scaled by multiplying by 0.1. Standard deviations, standardized skewness, covariance, and coskewness are computed on a monthly basis using deviations of daily averages from monthly averages, and reported results are averaged across all months in the sample. Note that the standard deviations in Panels B and C are not comparable with those in Panel A.

Panel A: Data description

	Mean	Std. dev.	Min	Max	Obs.
Spot price (€/MWh)	35.44	17.10	−130.09	200.04	55385
Demand (GWh/h)	59.72	10.88	32.81	86.41	54330
RES supply (GWh/h)	16.84	10.47	0.60	63.05	54330
Gas futures (€/MWh)	16.57	4.83	3.92	29.65	1600
Coal futures ($/MT)	67.39	17.26	38.45	102.60	1600
EUA futures (€/ton)	15.40	10.26	3.91	48.74	1599
Baseload futures (€/MWh)	37.05	9.47	17.02	64.27	1652
Peak load futures (€/MWh)	44.59	11.92	18.97	80.05	1652
Base forward premium (ex-post, €/MWh)	1.32	3.92	−7.82	11.32	76
Peak forward premium (ex-post, €/MWh)	1.52	4.98	−16.50	16.43	76

Panel B: Risk measures — baseload

Measure	Demand	RES supply	Residual demand	Spot price	Obs.
Std. dev.	5.57	5.73	8.83	8.44	76
Std. skew	−0.80	0.99	−0.47	−0.90	76
Cov(K, p)	–	−50.45	–	−50.45	76
Coskew(K^2, p) (×0.1)	–	−21.61	–	−21.61	76

Panel C: Risk measures — peak load

Measure	Demand	RES supply	Residual demand	Spot price	Obs.
Std. dev.	3.17	7.58	8.26	9.61	76
Std. skew	−1.16	0.51	−0.50	−0.43	76
Cov(K, p)	–	−62.23	–	−62.23	76
Coskew(K^2, p) (×0.1)	–	−24.94	–	−24.94	76

contracts. Column (1) presents the parameter estimates, including only the BL variables. The parameter estimate for variance has a negative sign, as predicted by BL, but is not significant. The parameter estimate for skewness is negatively significant, whereas BL predicted it to be positive. We explain this difference in sign from the convexity of the marginal production costs function that BL assume since their model does not include RES. As RES capacity increases over time and shifts the supply curve to the right, price spikes occur less frequently, but price dips may happen. Hence, the power industry hedges this downside risk, leading to a negative parameter.[35] Allowing for year fixed effects and fuel costs in column (2) does not change the results much.

Table 3: **Ex-ante Forward premium (Baseload)**

This table presents the parameter estimates for the ex-ante forward premium (equation 17) in baseload contracts. The results are obtained from daily data from January 2015 to April 2021. Columns (1) and (2) present the results for the BL model. Columns (3) and (4) show the results for the model derived in this paper. Columns (2) and (4) include year fixed effects and fuel costs as control variables. The magnitude of variance and coskewness are rescaled by 0.1, and the skewness is rescaled by 0.01. HAC standard errors are reported in parentheses.

	(1)	(2)	(3)	(4)
$\mathrm{Var}(p_S)$ ($\times 0.1$)	−0.01	−0.08	−0.45***	−0.48***
	(0.07)	(0.07)	(0.05)	(0.06)
$\mathrm{Skew}(p_S)$ ($\times 0.01$)	−0.05***	−0.05***	−0.07***	−0.08***
	(0.02)	(0.01)	(0.01)	(0.01)
$\mathrm{Cov}(K, p_S)$			−0.14***	−0.13***
			(0.01)	(0.01)
$\mathrm{Coskew}(K^2, p_S)$ ($\times 0.1$)			0.04***	0.05***
			(0.01)	(0.01)
Month FE	✓	✓	✓	✓
Year FE		✓		✓
Fuel costs		✓		✓
R^2	0.46	0.55	0.59	0.65
Adj. R^2	0.46	0.54	0.59	0.65
N	1579	1578	1579	1578

$^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

[35] Recall that we cannot predict the sign of the skewness parameter (β) in regression equation 17, because skewness preference always contributes a positive effect.

Table 4: **Ex-ante Forward Premium (Peak Load)**

This table presents the parameter estimates for the ex-ante forward premium (equation 17) in peak load contracts. The results are obtained by daily data from January 2015 to April 2021. Columns (1) and (2) present the results for the BL model. Columns (3) and (4) show the results for the model derived in this paper. Columns (2) and (4) include year fixed effects and fuel costs as control variables. The magnitude of variance and coskewness are rescaled by 0.1, and the skewness is rescaled by 0.01. HAC standard errors are reported in parentheses.

	(1)	(2)	(3)	(4)
$\mathrm{Var}(p_S)$ $(\times 0.1)$	-0.07^{*}	-0.14^{***}	-0.33^{***}	-0.33^{***}
	(0.04)	(0.04)	(0.05)	(0.05)
$\mathrm{Skew}(p_S)$ $(\times 0.01)$	-0.06^{***}	-0.06^{***}	-0.06^{***}	-0.05^{***}
	(0.01)	(0.01)	(0.01)	(0.01)
$\mathrm{Cov}(K, p_S)$			-0.10^{***}	-0.08^{***}
			(0.02)	(0.02)
$\mathrm{Coskew}(K^2, p_S)$ $(\times 0.1)$			0.02^{*}	0.02^{**}
			(0.01)	(0.01)
Month FE	✓	✓	✓	✓
Fuel costs		✓		✓
Year FE		✓		✓
R^2	0.58	0.67	0.64	0.71
Adj. R^2	0.58	0.67	0.64	0.70
N	1579	1578	1579	1578

$^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Columns (3) and (4) in Table 3 show the results when covariance and coskewness are included.[36] Our model predicts the parameter for covariance θ to be negative and the parameter for coskewness ξ to be positive. This is exactly what we observe in column (3), and both are significantly different from zero. Furthermore, the estimates for variance and skewness are both significantly negative. Especially the parameter estimate for variance has become more negative when we allow for covariance and skewness in the model. The estimate for skewness does not change. The model with covariance and coskewness fits the data better as the R^2's reported in columns (3) and (4) are higher than in (1) and (2) respectively. Allowing for the control variables improves the fit, but does not change the parameter estimates much.

Table 4 shows the results for the peak load contracts. Comparing with the base contracts, we

[36]Because the fuel cost data are missing for one trading day, the sample size is reduced by one when fuel costs are included as controls.

Table 5: **Ex-post Forward Premium (Baseload)**

This table presents the parameter estimates for the ex-post forward premium (equation 16) in base load contracts. The results are obtained by data observed on the last trading days of the months between January 2015 and April 2021. Columns (1) and (2) present the results for the BL model. Columns (3) and (4) show the results for the model derived in this paper. Columns (2) and (4) include month and year fixed effects and fuel costs as control variables. The magnitude of variance and coskewness are rescaled by 0.1, and the skewness is rescaled by 0.01. HAC standard errors are reported in parentheses.

	(1)	(2)	(3)	(4)
$\mathrm{Var}(p_S)$ $(\times 0.1)$	0.12	0.00	-0.68^{***}	-0.68^{***}
	(0.12)	(0.16)	(0.10)	(0.12)
$\mathrm{Skew}(p_S)$ $(\times 0.01)$	0.01	−0.04	-0.08^{**}	-0.08^{**}
	(0.06)	(0.06)	(0.04)	(0.04)
$\mathrm{Cov}(K, p_S)$			-0.17^{***}	-0.17^{***}
			(0.02)	(0.03)
$\mathrm{Coskew}(K^2, p_S)$ $(\times 0.1)$			0.06^{***}	0.04^{***}
			(0.02)	(0.01)
Month FE		✓		✓
Fuel costs		✓		✓
Year FE		✓		✓
R^2	0.04	0.42	0.38	0.59
Adj. R^2	0.02	0.19	0.34	0.41
N	76	76	76	76

$^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

observe similar results with the single difference being the parameter estimate for variance in the BL specification in columns (1) and (2) are significant. All signs are the same as for base load contracts and as predicted for covariance and coskewness. Adding those variables improves the fit. Allowing for control variables improves the fit further, but has no impact on the parameter estimates.

From observing the results in Tables 3 and 4, we conclude that all four factors explain the forward premium in power forward contracts. Beyond the variance and skewness of spot prices, as was predicted by BL, covariance and co-skewness significantly contribute to forward premiums as well.

We continue discussing the empirical results for the ex-post forward premiums. Tables 5 and 6 show the results for base and peak load contracts, respectively. Recall that for the ex-post results, we only use futures prices observed on the last trading day of the month. The spot prices and the forward premium factors are the realized values. First, we focus on the results for base load contracts in Table 5. The estimates of the parameters in the BL version of the model in column (1) are not significantly different from zero. When we include the covariance and co-skewness factors in column (3), all parameters are significant, and the signs are consistent with the model prediction. As with the ex-ante results, adding control variables does not change the parameter estimates much (although they improve the model's fit). The same results hold for the peak contracts in Table 6.

Table 6: **Ex-post Forward Premium (Peak Load)**

This table presents the parameter estimates for the ex-post forward premium (equation 16) in peak load contracts. The results are obtained by data observed on the last trading days of the months between January 2015 and April 2021. Columns (1) and (2) present the results for the BL model. Columns (3) and (4) show the results for the model derived in this paper. Columns (2) and (4) include month and year fixed effects and fuel costs as control variables. The magnitude of variance and coskewness are rescaled by 0.1, and the skewness is rescaled by 0.01. HAC standard errors are reported in parentheses.

	(1)	(2)	(3)	(4)
$\mathrm{Var}(p_S)$ $(\times 0.1)$	-0.07^{*}	-0.20^{***}	-0.42^{***}	-0.46^{***}
	(0.03)	(0.04)	(0.06)	(0.07)
$\mathrm{Skew}(p_S)$ $(\times 0.01)$	-0.03	-0.05^{**}	-0.04^{***}	-0.05^{***}
	(0.02)	(0.02)	(0.01)	(0.01)
$\mathrm{Cov}(K, p_S)$			-0.13^{***}	-0.12^{***}
			(0.03)	(0.03)
$\mathrm{Coskew}(K^2, p_S)$ $(\times 0.1)$			0.06^{***}	0.04^{**}
			(0.02)	(0.01)
Month FE		✓		✓
Fuel costs		✓		✓
Year FE		✓		✓
R^2	0.08	0.51	0.40	0.64
Adj. R^2	0.05	0.32	0.37	0.49
N	76	76	76	76

$^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The results show that the risk factors proposed by our model can explain the forward premiums observed in German futures contracts. These factors improve the fit of the models compared to models that include only the BL factors. Where demand uncertainty was the only source of risk in the BL model, we include RES supply uncertainty as an additional source of uncertainty that CES producers and retailers have to deal with. The results in this paper show that the risk caused by this uncertainty is priced.

5 Concluding remarks

In this paper, we propose an equilibrium model for power forward prices. Our starting point is that existing models, such as Bessembinder and Lemmon (2002) that assume power is produced by non-intermittent conventional energy sources (CES) and agents have mean–variance preferences, do not match the data. In recent years, power markets have seen a rapid increase in supply from renewable energy sources (RES), such as wind and solar. This motivates us to extend the model by incorporating RES supply. Unlike CES, RES production is intermittent and has zero marginal costs. As a result, spot prices are lower but more volatile. Moreover, the distribution of RES supply is often skewed, which induces skewness in power spot prices and consequently in profits. To capture how agents respond to this profit skewness, we introduce mean–variance–skewness preferences.

In addition to spot price variance and skewness found by Bessembinder and Lemmon (2002), our equilibrium model proposes new factors related to covariance and coskewness between RES supply and spot prices that can help explain the power forward premium. We provide empirical support for these new factors and their predicted signs explaining the forward premiums observed in the German power market.

Moreover, our model predicts that the sign of skewness, which Bessembinder and Lemmon (2002) predict to be positive, cannot be determined as it depends on the convexity of the supply curve. The low marginal production costs of RES shift the supply curve to the right, such that demand is supplied from a flatter region of the supply curve. Using data from Germany, we find that the skewness parameter is negative. Our model shows that the sign of the variance parameter is also indefinite. To our knowledge, this provides a first theoretical explanation for the mixed empirical findings in the literature.

One could argue that the validity of our model is limited as it is based on a market setting where power storage capacity is limited and/or power demand is price-inelastic in the short term. Once power markets have sufficiently large storage capacity and agents change their consumption quantities in response to changes from RES supply, the variance and skewness of spot prices and the relationship between supply from RES and spot prices will eventually disappear. As a consequence, the importance of the factors explaining the power forward premium disappears, and forward prices will behave more in line with what is predicted by the theory of storage. We think that this will not happen soon, as it requires huge investments and behavior changes, specifically for the medium-term storage. Until then, we believe that our model provides a better understanding of how agents in power markets hedge risk and what explains forward premiums in markets where renewable energy sources produce a growing volume of electricity.

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Appendix

A Distribution of spot prices

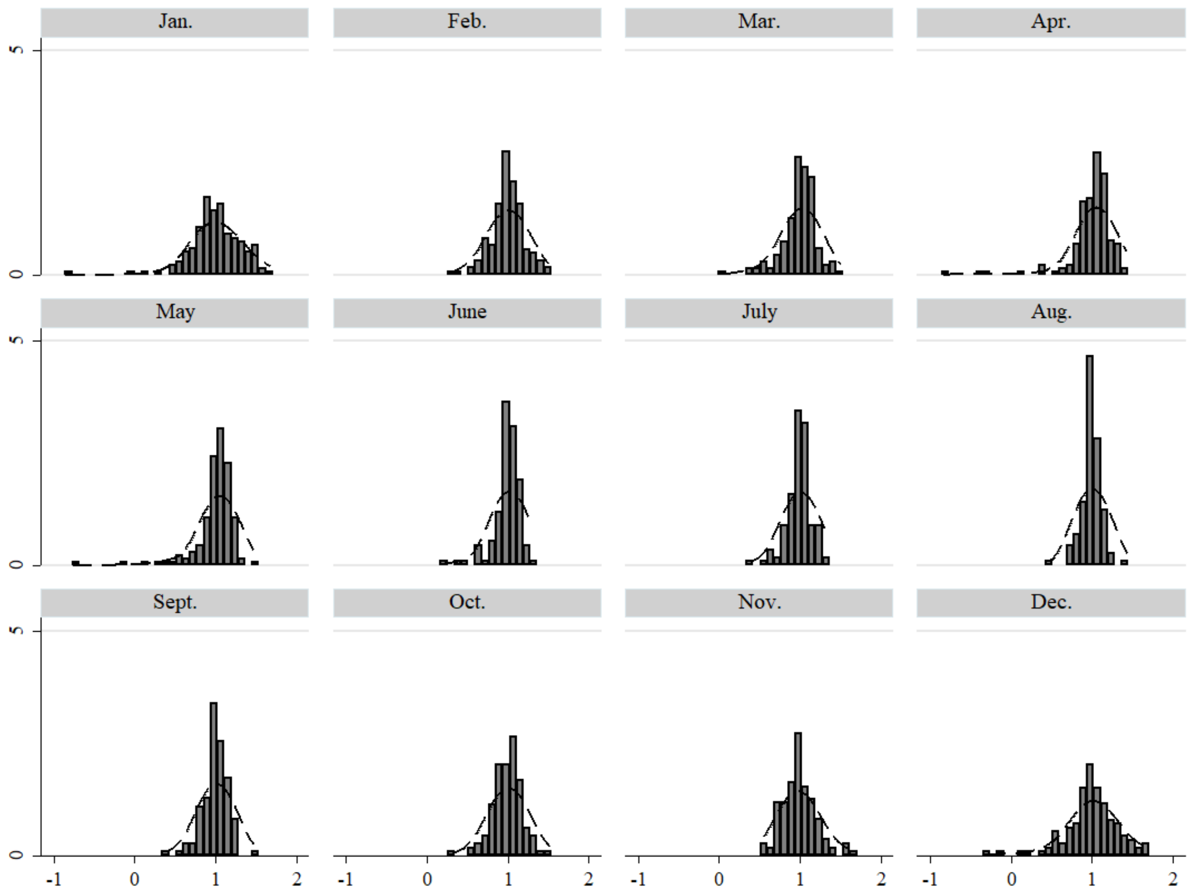

Figure 1: **Distribution of Spot Prices.**

This figure shows the empirical distribution of electricity spot prices across months in Germany. The sample covers the period from January 2015 to April 2021.

B Distribution of RES supply

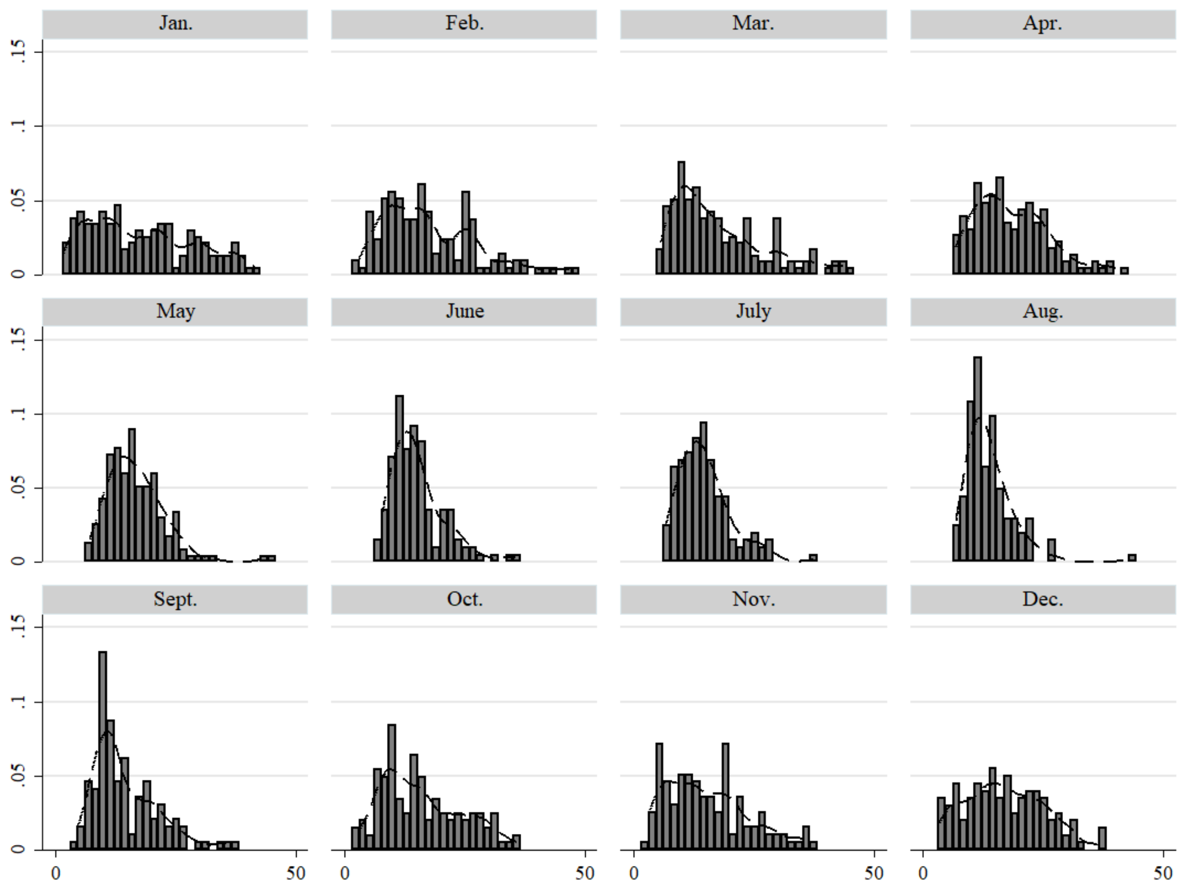

Figure 2: **Distribution of RES Supply.**

This figure shows the empirical distribution of power supply from renewable energy sources (RES) across months in Germany. The sample covers the period from January 2015 to April 2021.

C Proof of equilibrium forward pricing

Substituting the derivative of skewness into the first-order condition for forward positions (see Eqs. 7 and 8) yields a quadratic equation in $Q^F_{I(J)}$:

$$\frac{\text{Skew}(p_S)}{\text{Var}(p_S)}\,(Q^F_{I(J)})^2 - 2\,\tau_{I(J)}\,Q^F_{I(J)} + \frac{\phi}{\text{Var}(p_S)}\,(p_{I(J)} - p_F) = 0, \tag{C.1}$$

where

$$\tau_{I(J)} = \frac{\text{Coskew}(\rho_{I(J)}, p_S^2)}{\text{Var}(p_S)} - \frac{\phi}{2\gamma},$$
$$p_{I(J)} = \mathbb{E}(p_S) - \frac{1}{\gamma}\text{Cov}(\rho_{I(J)}, p_S) + \frac{1}{\phi}\text{Coskew}(\rho_{I(J)}^2, p_S).$$

Existence Equation (C.1) admits real solutions only if the discriminant is non-negative:

$$\Delta = \tau_{I(J)}^2 - \phi \cdot \frac{\text{Skew}(p_S)}{\text{Var}(p_S)^2}\,(p_{I(J)} - p_F) \;\geq\; 0. \tag{C.2}$$

Roots When $\Delta \geq 0$, the quadratic admits two algebraic solutions:

$$Q_{I(J),\pm}^F = \frac{\tau_{I(J)} \;\pm\; \sqrt{\tau_{I(J)}^2 - \phi \cdot \frac{\text{Skew}(p_S)}{\text{Var}(p_S)^2}\,(p_{I(J)} - p_F)}}{\text{Skew}(p_S)/\text{Var}(p_S)}.$$

Second-order condition The second derivative of expected utility with respect to the forward position is

$$U''(Q_{I(J)}^F) = \tau_{I(J)} - \frac{\text{Skew}(p_S)}{\text{Var}(p_S)}\,Q_{I(J)}^F.$$

The second-order condition changes sign at the threshold forward position

$$Q_{I(J)}^{F,\text{crit}} = \frac{\text{Var}(p_S)}{\text{Skew}(p_S)}\tau_{I(J)}.$$

We can rewrite the second derivative as:

$$U''(Q_{I(J)}^F) = -\frac{\text{Skew}(p_S)}{\text{Var}(p_S)}\,(Q_{I(J)}^F - Q_{I(J)}^{F,\text{crit}}).$$

- If $\text{Skew}(p_S) > 0$, the SOC requires $Q_{I(J)}^F > Q_{I(J)}^{F,\text{crit}}$. The larger root $Q_{I(J),+}^F$ satisfies this condition and is the maximizer; the smaller root is a minimizer.
- If $\text{Skew}(p_S) < 0$, the SOC requires $Q_{I(J)}^F < Q_{I(J)}^{F,\text{crit}}$. The smaller root $Q_{I(J),-}^F$ satisfies this condition and is the maximizer; the larger root is a minimizer.
- If $\text{Skew}(p_S) = 0$, Eq. (C.1) reduces to a linear equation with a unique solution, which

automatically satisfies the SOC.

Hence, the solution that maximizes the agent's utility is:

$$Q^F_{I(J)} = \frac{\tau_{I(J)} + \sqrt{\tau^2_{I(J)} - \phi \cdot \frac{\text{Skew}(p_S)}{\text{Var}(p_S)^2}\,(p_{I(J)} - p_F)}}{\text{Skew}(p_S)/\text{Var}(p_S)}. \tag{C.3}$$

The market-clearing forward price should be determined such that the sum of forward positions across conventional producers and retailers is zero.

$$Q^F_I + Q^F_J = 0$$

Combining the optimal-position expressions with the market-clearing condition yields the following cases. If $\tau_I + \tau_J < 0$, the equilibrium forward price is

$$p_F = \frac{\tau_I p_J + \tau_J p_I}{\tau_I + \tau_J} + \frac{\phi}{4} \frac{\text{Skew}(p_S)}{\text{Var}(p_S)^2} \frac{(p_I - p_J)^2}{(\tau_I + \tau_J)^2}. \tag{C.4}$$

If $\tau_I + \tau_J > 0$, a market equilibrium does not exist. The boundary case $\tau_I + \tau_J = 0$ is degenerate: the candidate forward price and forward positions become unbounded as $\tau_I + \tau_J$ approaches zero. Substituting equations (C.4) into equation (C.2) shows that the non-negative condition holds.

D Proof of reduced-form equilibrium

Taylor series says:

$$f(x) \approx f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2}(x - x_0)^2$$

Hence, to approximate p_S^x around $\mathbb{E}(p_S)$ gives:

$$\begin{aligned} p_S^x &\approx \mathbb{E}(p_S)^x + x[\mathbb{E}(p_S)]^{x-1}(p_S - \mathbb{E}(p_S)) + \frac{x(x-1)[\mathbb{E}(p_S)]^{x-2}}{2}(p_S - \mathbb{E}(p_S))^2 \\ &= [\mathbb{E}(p_S)]^x \left[1 - x + \frac{x(x-1)}{2}\right] + x(2-x)[\mathbb{E}(p_S)]^{x-1} p_S + \frac{x(x-1)}{2}[\mathbb{E}(p_S)]^{x-2} p_S^2 \end{aligned}$$

Similarly,

$$p_S^{x+1} = [\mathbb{E}(p_S)]^{x+1}\left[-x + \frac{x(x+1)}{2}\right] - (x+1)(x-1)[\mathbb{E}(p_S)]^x p_S + \frac{x(x+1)}{2}[\mathbb{E}(p_S)]^{x-1} p_S^2$$

For simplification, we define:

$$\begin{aligned}
\kappa_1 &= -(x+1)(x-1)[\mathbb{E}(p_S)]^x, \\
\kappa_2 &= \frac{x(x+1)}{2}[\mathbb{E}(p_S)]^{x-1}, \\
\psi &= p_R x(2-x)[\mathbb{E}(p_S)]^{x-1} + (x+1)(x-1)[\mathbb{E}(p_S)]^x.
\end{aligned}$$

Then, we can rewrite the covariance between ρ_I and p_S as

$$\begin{aligned}
\mathrm{Cov}(\rho_I, p_S) &= \frac{1}{(x+1)a^x}\mathrm{Cov}(p_S^{x+1}, p_S) \\
&= \frac{\kappa_1 + 2\kappa_2\mathbb{E}(p_S)}{(x+1)a^x}\mathrm{Var}(p_S) + \frac{\kappa_2}{(x+1)a^x}\mathrm{Skew}(p_S).
\end{aligned}$$

For coskewness, we keep only up to third-order central and mixed moments.

$$\begin{aligned}
\mathrm{Coskew}(\rho_I^2, p_S) &= \frac{1}{(x+1)^2 a^{2x}}\mathrm{Coskew}(p_S^{x+1}, p_S^{x+1}, p_S) \\
&= \frac{\kappa_1^2}{(x+1)^2 a^{2x}}\mathrm{Skew}(p_S).
\end{aligned}$$

Hence,

$$\begin{aligned}
p_I &= \mathbb{E}(p_S) - \frac{1}{\gamma}\mathrm{Cov}(\rho_I, p_S) + \frac{1}{\phi}\mathrm{Coskew}(\rho_I^2, p_S) \\
&= \mathbb{E}(p_S) - \frac{\mathbb{E}(p_S)^x}{\gamma a^x}\mathrm{Var}(p_S) + \left[\frac{(x-1)^2[\mathbb{E}(p_S)]^{2x}}{\phi a^{2x}} - \frac{x[\mathbb{E}(p_S)]^{x-1}}{2\gamma a^x}\right]\mathrm{Skew}(p_S).
\end{aligned}$$

Similarly, we get

$$\begin{aligned}
p_J = {} & \mathbb{E}(p_S) \\
& - \left[\frac{[\mathbb{E}(p_S)]^{x-1}}{\gamma a^x} \big(x p_R - (x+1)\mathbb{E}(p_S) \big) - \frac{\mathbb{E}(K)}{\gamma} \right] \operatorname{Var}(p_S) \\
& + \left[\frac{\psi^2}{\phi a^{2x}} - \frac{x[\mathbb{E}(p_S)]^{x-2}}{2\gamma a^x} \big((x-1) p_R - (x+1)\mathbb{E}(p_S) \big) \right] \operatorname{Skew}(p_S) \\
& - \left[\frac{p_R - \mathbb{E}(p_S)}{\gamma} \right] \operatorname{Cov}(K, p_S) \\
& + \left[\frac{p_R^2}{\phi} \right] \operatorname{Coskew}(K^2, p_S) \\
& + \left[\frac{2\psi p_R}{a^x \phi} + \frac{1}{\gamma} \right] \operatorname{Coskew}(K, p_S^2).
\end{aligned}$$

Substituting the results into equation (12) and rearranging the terms give the reduced-form representation as shown in equation (15).